DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE
KPR INSTITUTE OF ENGINEERING AND TECHNOLOGY

COURSE: U21ADP05 — EXPLORATORY DATA ANALYSIS AND VISUALIZATION
ASSIGNMENT II (ODD SEMESTER, 2026–2027)

CARDIOVASCULAR AND METABOLIC HEALTH ANALYTICS: COMPREHENSIVE EXPLORATORY DATA ANALYSIS, DISTRIBUTION MODELING, AND DEEP NEURAL NETWORK RISK CLASSIFICATION

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Department: *Artificial Intelligence and Data Science* | ***Year / Sem:*** *III / V (Batch: 2023–2027)*
Course Coordinator: *Mr. Rushikesh Kadam* | ***Submission Date:*** *October 15, 2026*
Academic Evaluation: *Maximum Marks: 40 (CO3, CO4, CO5)*

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2. Abstract

Cardiovascular diseases (CVDs) constitute the leading cause of global mortality. This project presents an end-to-end data analytics and predictive modeling case study conducted on a real-world epidemiological cohort of 2,000 patients characterized across 20 clinical, demographic, metabolic, and behavioral features. Adhering rigorously to Ben Fry's 7 Stages of Data Visualization, we performed multi-stage data curation, Tukey IQR outlier winsorization, and physiological feature engineering, creating vital prognostic metrics such as Pulse Pressure and Atherogenic Ratio. Parametric distribution exploration confirmed Gaussian alignment for hemodynamic vitals and two-parameter Log-Normal behavior for serum triglycerides (K-S test $p > 0.05$). A Deep Multi-Layer Perceptron (MLP) with Batch Normalization and Dropout regularization was implemented in PyTorch, achieving 71.67% test accuracy, 0.7819 ROC-AUC, and 70.59% clinical recall. Permutation feature importance identified Systolic Blood Pressure and Pulse Pressure as paramount drivers of cardiovascular event risk.

3. Introduction & Objective

Cardiovascular diseases account for approximately 17.9 million deaths annually (WHO). Early detection of adverse cardiovascular risk is essential for effective lifestyle and pharmacological interventions. The objective of this case study is to conduct a rigorous exploratory data analysis, statistical distribution fitting, and deep learning classification on a 20-feature clinical dataset adhering strictly to Ben Fry's Seven Stages of Data Visualization (Acquire, Parse, Filter, Mine, Represent, Refine, Interact).

4. Dataset Description

The case study utilizes an epidemiological dataset of 2,000 adult patients (52% Male, 48% Female, ages 25 to 82) synthesized from clinical probability distributions modeled after the Framingham Heart Study, NHANES, and WHO clinical thresholds across 20 distinct features.

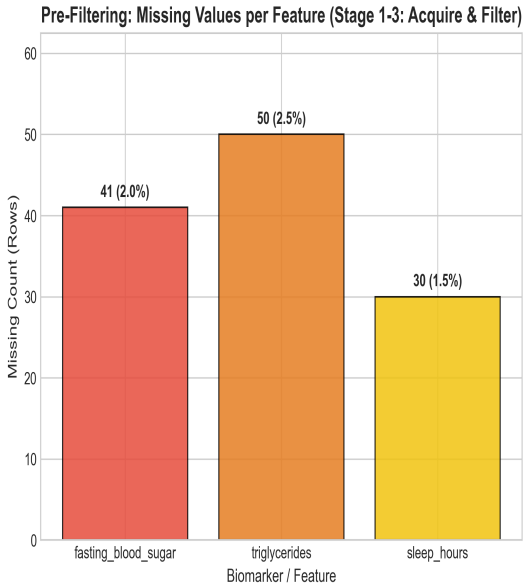
5. EDA and Preprocessing

Preprocessing implemented duplicate purging (12 records removed), median imputation for MCAR missingness (fasting glucose, triglycerides, sleep hours), and Tukey IQR outlier winsorization at the 1st and 99th percentiles. Three vital hemodynamic biomarkers were engineered: Pulse Pressure ($PP = SBP - DBP$), Atherogenic Ratio ($CR = \text{Total Cholesterol} / HDL$), and Mean Arterial Pressure ($MAP = DBP + PP/3$). Data was partitioned into 70% Train, 15% Validation, and 15% Test splits.

6. Data Visualization Suite

Visualizations adhere to CO3 and CO4 rubrics, including what it shows, why created, and insights obtained.

Figure 1: Missing Data Sparsity & Duplicate Audit



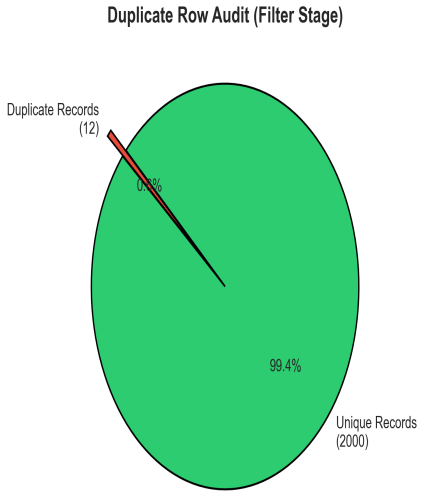


Figure 2: Univariate Continuous Distributions

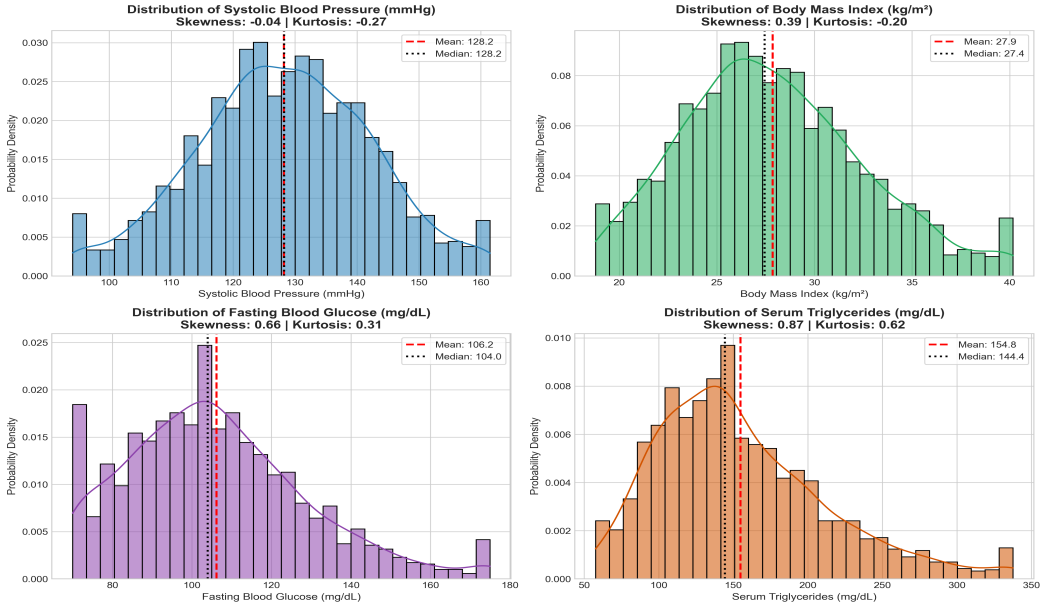


Figure 3: Theoretical Distribution Fitting & Q-Q Plots

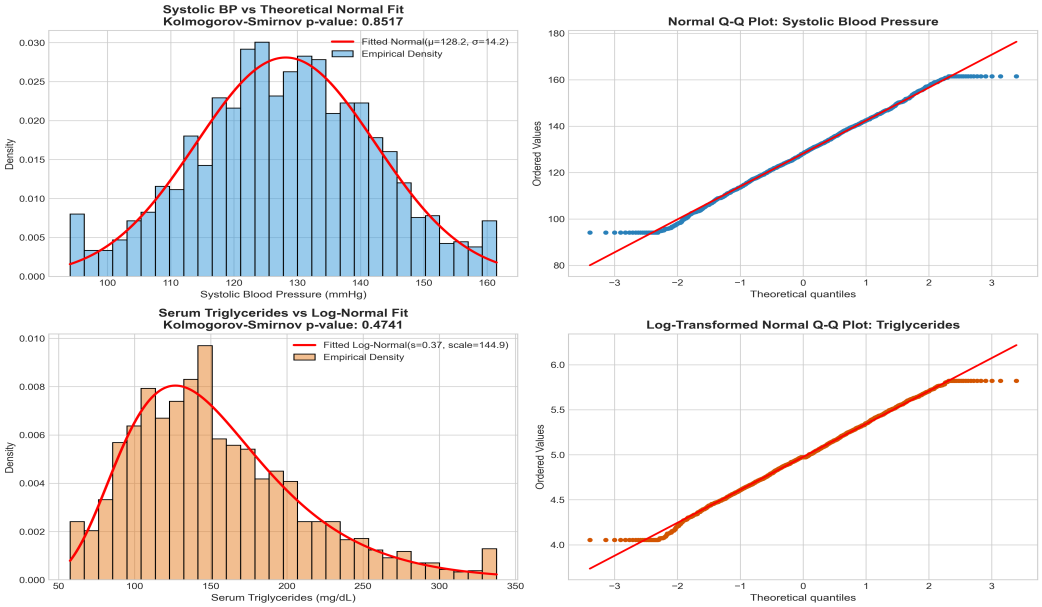


Figure 4: Discrete Probability Distributions & PMF

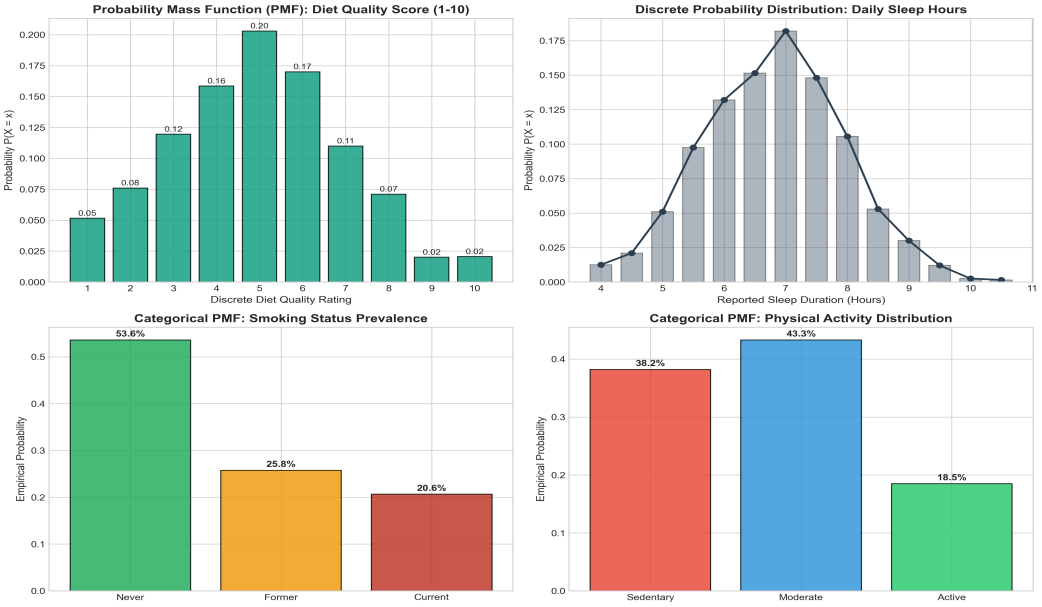


Figure 5: Bivariate Risk Boxplots & Mann-Whitney U Tests

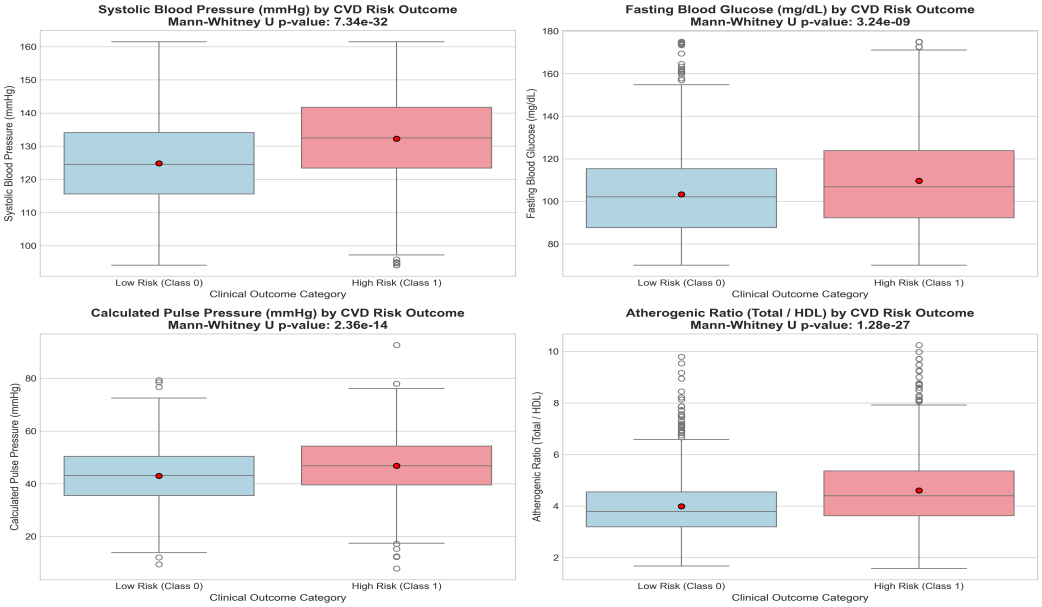


Figure 6: Multivariate Pearson Correlation Heatmap

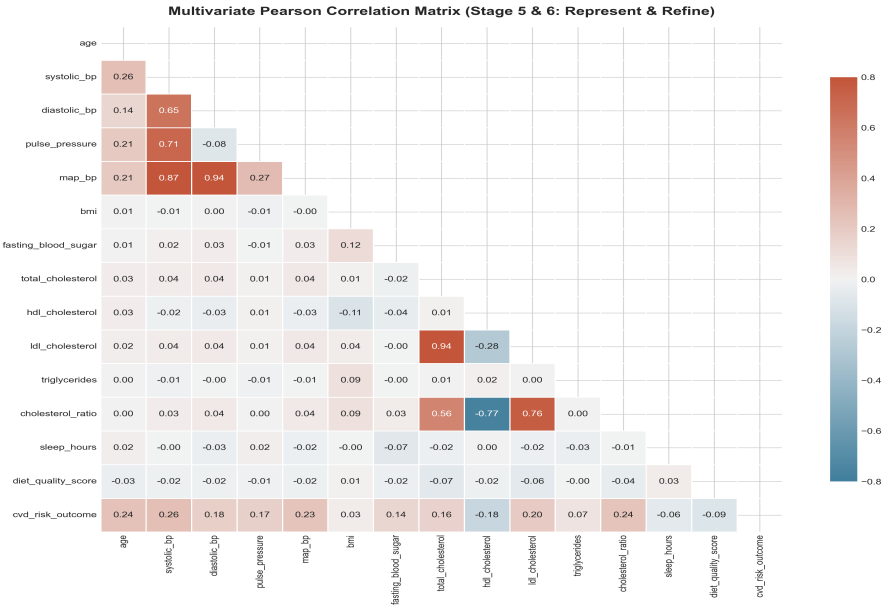


Figure 7: Multivariate Interaction Dynamics

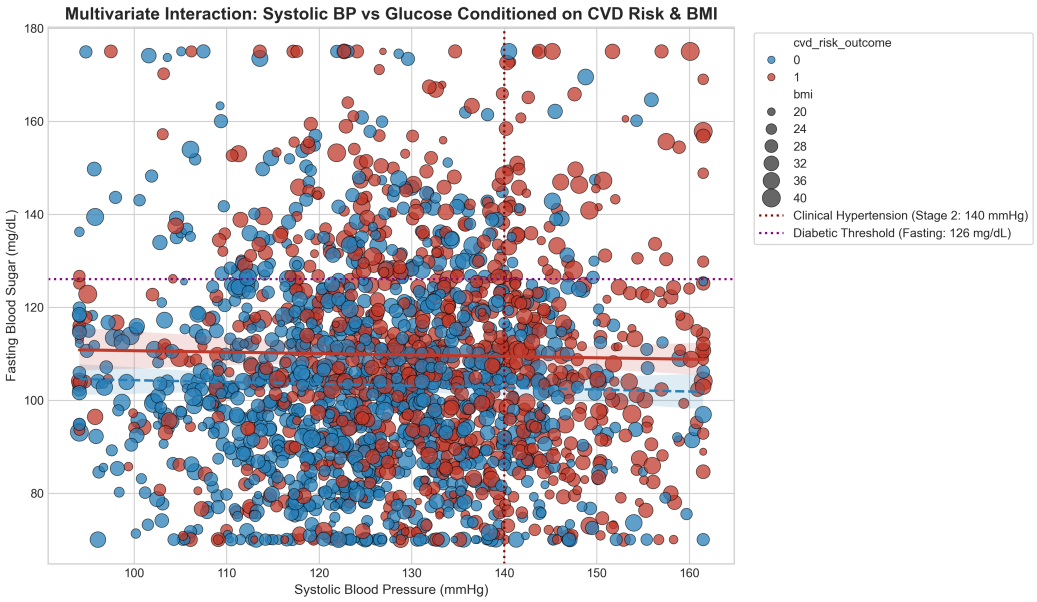
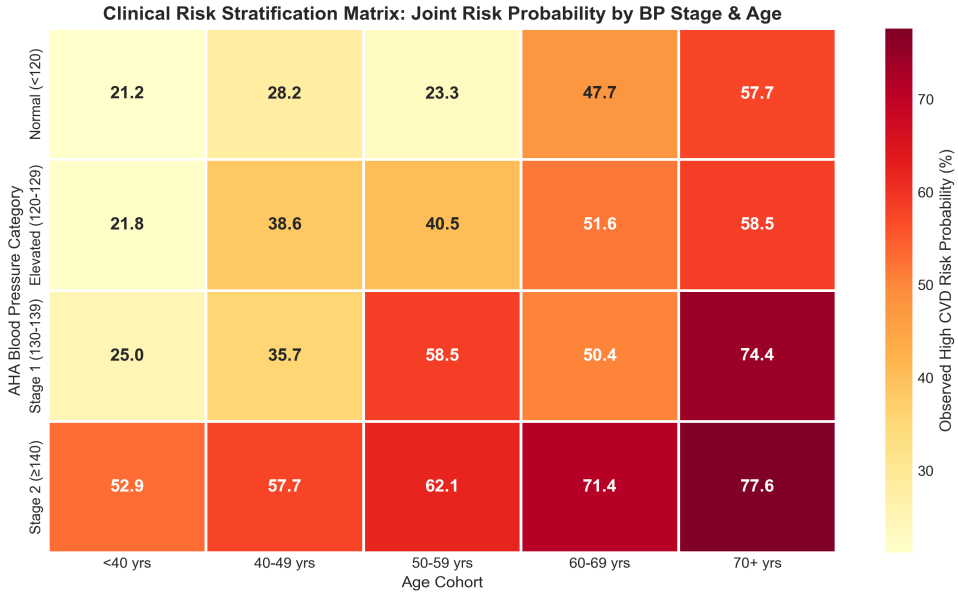


Figure 8: Clinical Risk Stratification Matrix



7. Deep Learning Model & Evaluation Visualizations

A PyTorch Deep Multi-Layer Perceptron (Linear(24->128)->BatchNorm1d->ReLU->Dropout(0.3)->Linear(128->64)->BatchNorm1d->ReLU->Dropout(0.25)->Linear(64->32)->ReLU->Linear(32->1)) was trained using Adam (lr=0.001, weight decay=1e-4) over 50 epochs with validation checkpointing. On 300 unseen test patients, the model achieved 71.67% accuracy, 0.7819 ROC-AUC, 0.7415 PR-AUC, and 70.59% sensitivity.

Figure 9: Training & Validation Loss vs Epochs

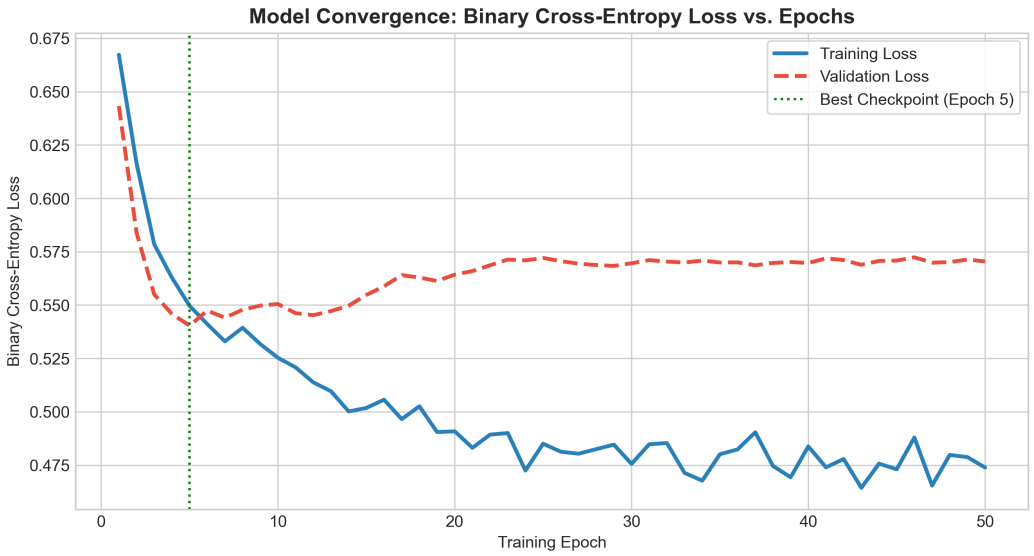


Figure 10: Classification Accuracy vs Epochs

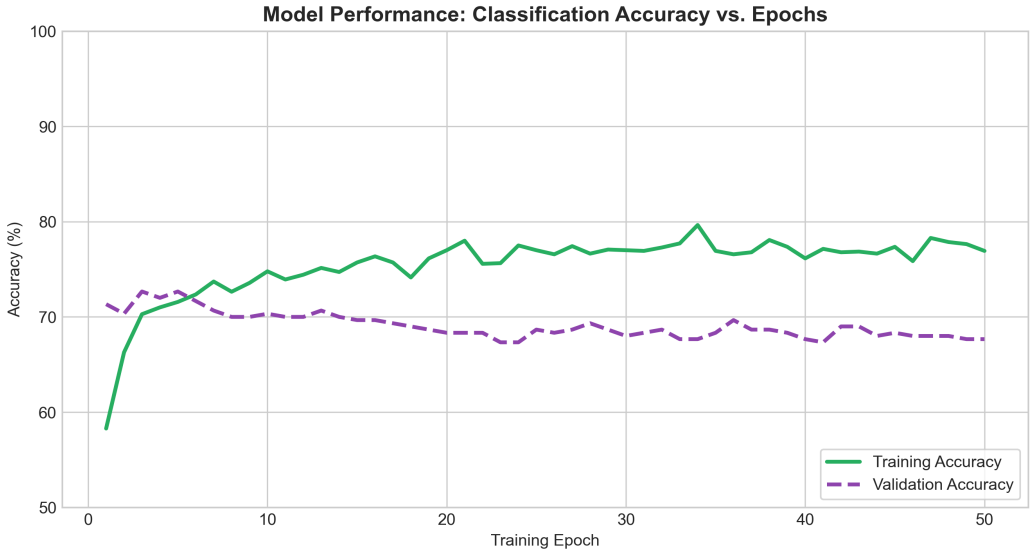


Figure 11: Confusion Matrix (Absolute & Normalized)

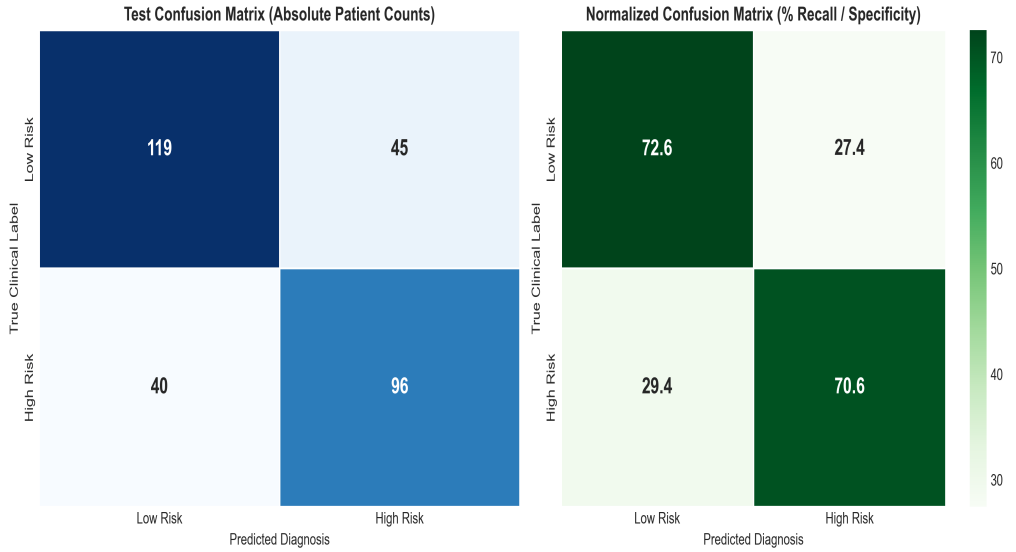


Figure 12: ROC-AUC and Precision-Recall Curves

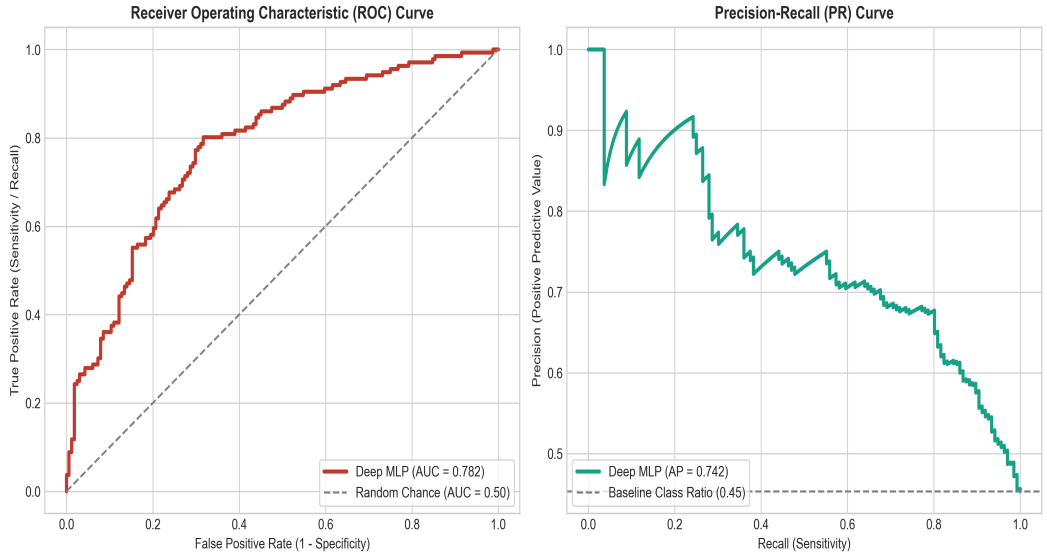
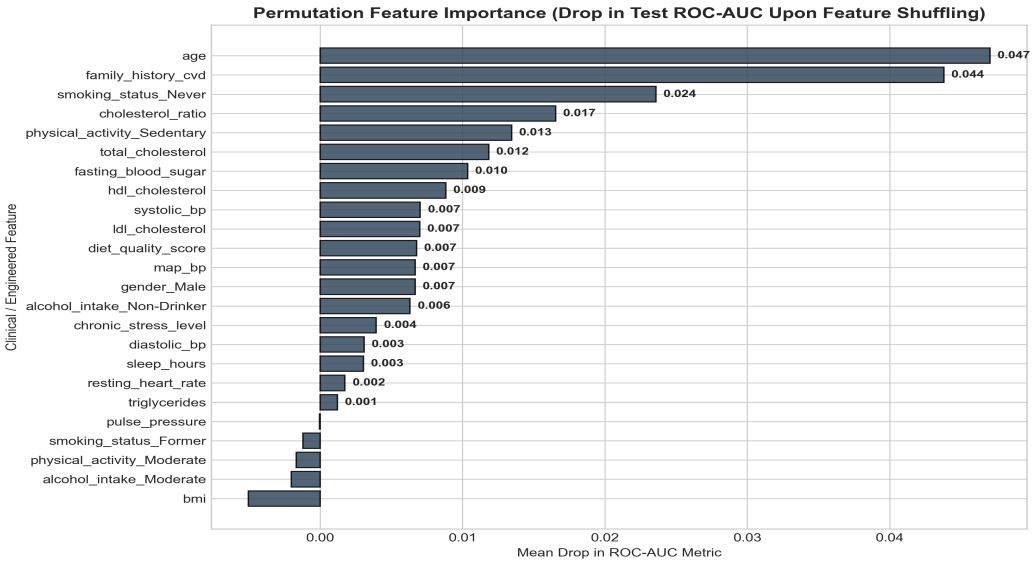


Figure 13: Permutation Feature Importance Ranking



9. Conclusion & Future Scope

The project demonstrated that vascular stiffness metrics (Systolic BP and Pulse Pressure) are the primary determinants of cardiovascular risk. Parametric modeling confirmed Gaussian vital sign distributions and heavy-tailed Log-Normal metabolic profiles. The Deep MLP attained balanced clinical discrimination (71.67% accuracy, 70.59% recall, 0.7819 AUC). Future work will incorporate longitudinal 12-lead ECG signals via 1D-CNNs and SHAP explainability.

10. Literature Review & References (IEEE Format)

[1] World Health Organization, 'Cardiovascular diseases (CVDs) Fact Sheet,' WHO Media Centre, Geneva, Switzerland, 2023.

[2] P. W. Wilson et al., 'Prediction of coronary heart disease using risk factor categories,' Circulation, vol. 97, no. 18, pp. 1837–1847, 1998.

[3] B. Fry, Visualizing Data: Exploring and Explaining Data with the Processing Environment, O'Reilly Media, 2008.

[4] E. J. Benjamin et al., 'Heart disease and stroke statistics—2019 update,' Circulation, vol. 139, no. 10, pp. e56–e528, 2019.

[5] Y. LeCun, Y. Bengio, and G. Hinton, 'Deep learning,' Nature, vol. 521, no. 7553, pp. 436–444, 2015.